

Cournot Duopoly

$$p = a - bQ, \quad a, b > 0$$

$$Q = q_1 + q_2, \quad q_1, q_2 \geq 0$$

$$\pi_1(q_1, q_2) = TR - TC = pq_1 - cq_1 \quad \text{--- (1)}$$
$$= (a - b(q_1 + q_2))q_1 - cq_1$$

$$\pi_2(q_1, q_2) = (a - b(q_1 + q_2))q_2 - cq_2 \quad \text{--- (2)}$$

Best-response f^o

Firm 1:- FOC: $\frac{\partial \pi_1}{\partial q_1}(q_1, q_2) = 0$

$$\Rightarrow \frac{\partial \pi_1}{\partial q_1} = a - b(q_1 + q_2) - bq_1 - c = 0$$
$$\Rightarrow (a - c) - 2bq_1 - bq_2 = 0 \quad \text{--- (3)}$$

Best-response fⁿ

Firm 2:-

$$\frac{\partial \pi_2(q_1, q_2)}{\partial q_2} = 0$$

$$\Rightarrow \frac{\partial \pi_2}{\partial q_2} = a - b(q_1 + q_2) - bq_2 - c = 0 \quad \text{--- (4)}$$

From (3) & (4),

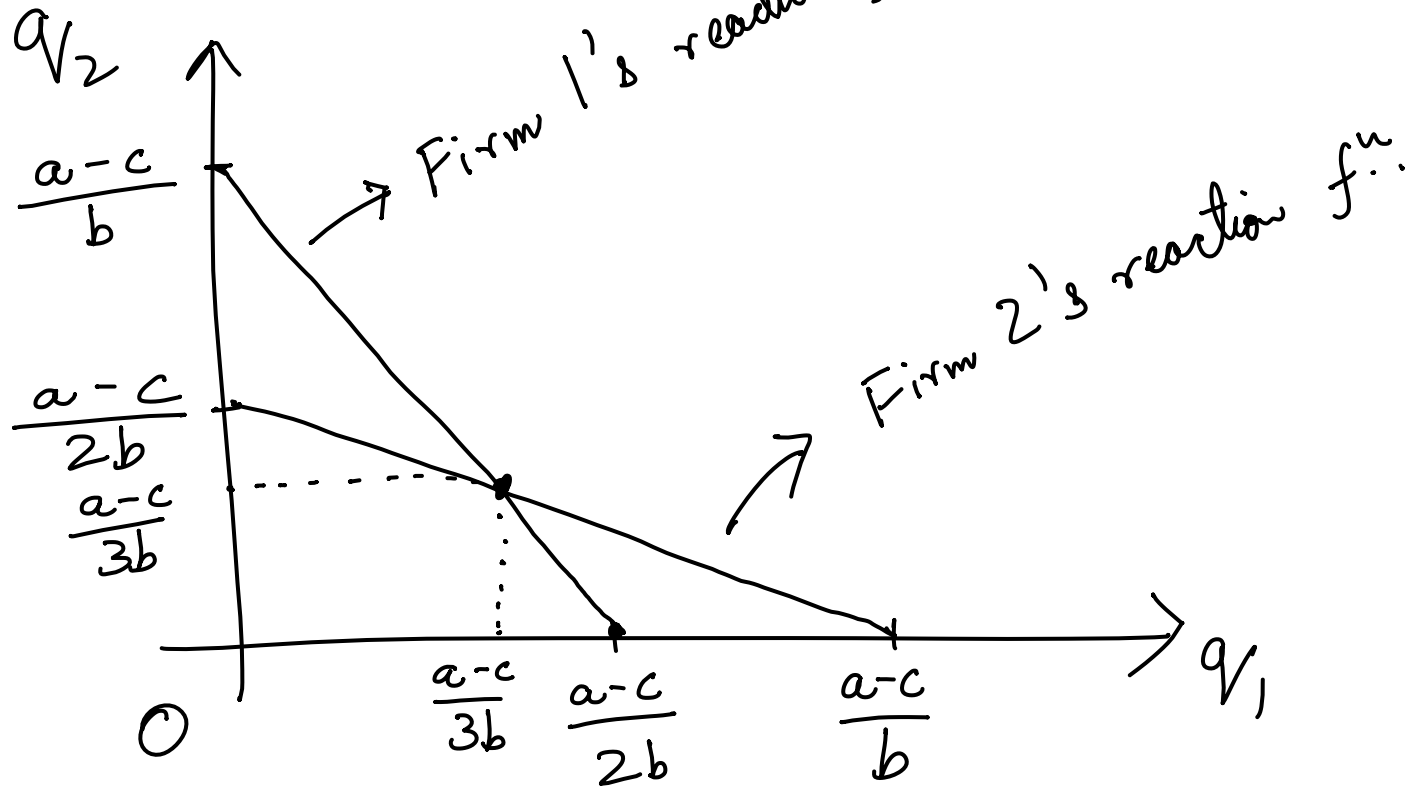
$$2bq_1 + bq_2 = a - c \quad \checkmark \quad \text{--- (5)}$$

$$bq_1 + 2bq_2 = a - c \quad \text{--- (6)}$$

Reaction / Best response f^u



Best-response / Reaction fⁿ.



From ⑤ & ⑥,

$$q_1^* = \frac{a-c}{3b}, \quad q_2^* = \frac{a-c}{3b}$$

(q_1^*, q_2^*) is the Cournot - Nash Equilibrium

Duopoly
in profits
Cournot - Nash
equilibrium

$$\left[\begin{aligned} \pi_1^*(q_1^*, q_2^*) &= \left(a - 2b \frac{a-c}{3b} \right) \frac{a-c}{3b} - c \frac{a-c}{3b} \\ &= \frac{(a-c)^2}{9b} \\ \pi_2^*(q_1^*, q_2^*) &= \frac{(a-c)^2}{9b} \end{aligned} \right]$$

Monopoly problem

$$\begin{aligned}\pi(Q) &= TR - TC \\ &= pQ - cQ \\ &= (a - bQ)Q - cQ \\ &= aQ - bQ^2 - cQ\end{aligned}$$

FOC

$$\frac{\partial \pi}{\partial Q} = 0$$

$$\Rightarrow Q^* = \frac{a - c}{2b}$$

Each firm produces $\frac{a - c}{4b}$

Under-monopoly,

$$\pi^*(Q^*) = \left(a - b \frac{a-c}{2b} \right) \frac{a-c}{2b} - c \left(\frac{a-c}{2b} \right)$$

$$= \frac{(a-c)^2}{4b}$$

Assuming profits are equally shared, each firm's profit is $\frac{(a-c)^2}{8b}$ under monopoly.

That is greater than $\frac{(a-c)^2}{9b}$ that each firm earns as profit in Cournot-Nash equilib^u.